Problem Set 9

A. Oguri and Polygons

tianzheyu submissions

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#	When	Who	Problem	Lang	Verdict	Time	Memory
386025426	Aug/07/2026 16:16 UTC+10	tianzheyu	A - Oguri and Polygons	C++23 (GCC 14-64, msys2)	Accepted	1281 ms	25200 KB

Process

Precomputed the number of points strictly inside every possible triangle formed by the given vertices using bitsets. For each query, I triangulated the given convex polygon and summed the precomputed triangle values to get the total points inside.

Challenges and Overcoming

When implementing the initial bitset solution, I ran into a performance bottleneck where I was computing the intersection of large bitsets for every single query. I found it when my submission hit a Time Limit Exceeded on Test 14, realizing that even highly optimized bitset operations were too slow for the 400,000 queries. Next time to prevent this performance issue, I will explicitly look for unusually small constraints in the problem description (like $N \le 40$) before coding and use them to shift the heavy computation entirely into an $O(N^3)$ precomputation phase, guaranteeing $O(K)$ minimal work per query.